

Cyrus Guided Notes: MIT Underactuated Robotics

MIT course pack

2026-05-15

Course source

Official source: <https://underactuated.mit.edu/>

Why this course is in the system. Entry point for robot dynamics, optimal control, trajectories, stability, and nonlinear systems.

Learning output. Build inverted-pendulum, phase-portrait, and trajectory-optimization interactions on the web.

Prerequisite checkpoint

- Dynamics
- Optimization
- State-space control

Formula checkpoint

Do not memorize these first. Read each formula as a sentence: what changes, what is observed, what is optimized, and what output it produces.

$$M(\mathbf{q})\ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = B\mathbf{u}$$

$$J = \sum_{k=0}^N \ell(\mathbf{x}_k, \mathbf{u}_k)$$

$$V(\mathbf{x}) > 0, \quad \dot{V}(\mathbf{x}) < 0$$

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k)$$

Visual page

Use an inverted pendulum phase portrait to show underactuation and stabilization.

GoodNotes pages

- Page MIT-R001: underactuated dynamics
- Page MIT-R002: trajectory optimization

Web interaction

A pendulum slider for angle and control effort.

Obsidian and Notion

Layer	Output
Obsidian	MIT Robotics -> underactuated robotics
Notion	Robot-control concepts and experiment state.

First study move

Open the official source, copy the prerequisite checkpoint into GoodNotes, then write one formula in your own words before watching or reading more.